UNIVERSITY OF SCIENCE AND TECHNOLOGY OF SOUTHERN PHILIPPINES

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**UNIVERSITY OF SCIENCE AND TECHNOLOGY OF SOUTHERN PHILIPPINES**

# System Project:
# Student Record Management System

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IT323 - Midterm PIT

Section: IT3R10

Bachelor of Science in Information Technology

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## SYSTEM PROJECT: Student Record Management System Documentation

## ER Diagram

| | students | |
|---|---|---|
| PK | id | INT AUTO_INCREMENT |
| UQ | student_number | VARCHAR(20) NOT NULL |
| | first_name | VARCHAR(100) NOT NULL |
| | last_name | VARCHAR(100) NOT NULL |
| | course | VARCHAR(100) NOT NULL |
| | year_level | INT NOT NULL |
| | created_at | TIMESTAMP DEFAULT CURRENT_TIMESTAMP |

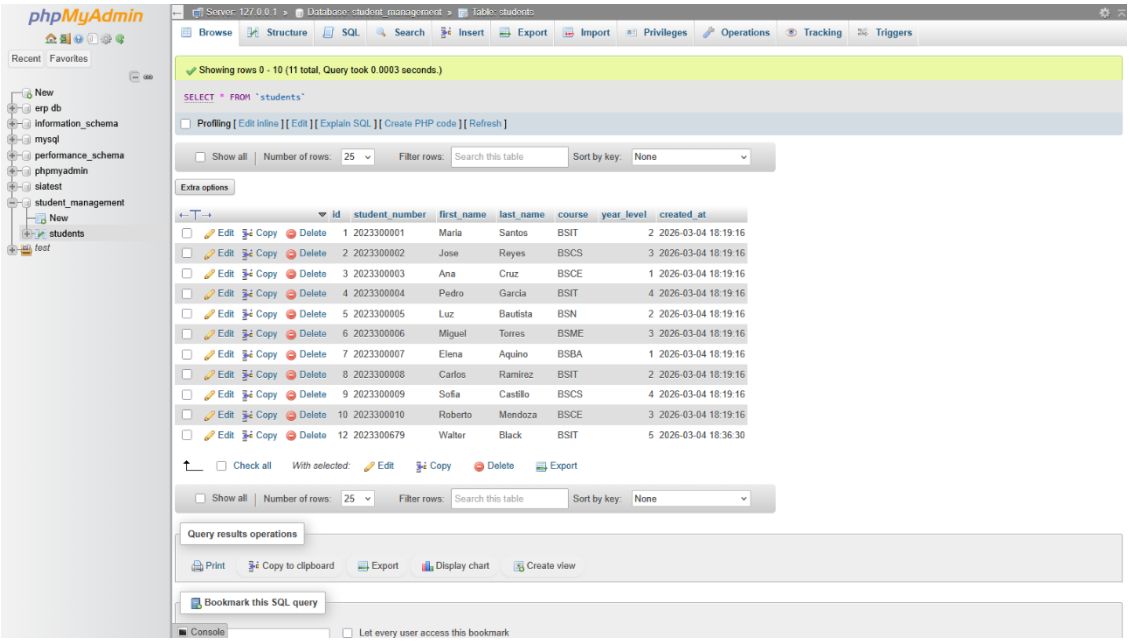
Legend:
PK = Primary Key (Unique identifier for each record)
UQ = Unique Constraint (No duplicates allowed)

## API Documentation

| Method | Endpoint | Description | Status Codes |
|---|---|---|---|
| POST | /api/students | Add new student | 201, 400, 500 |
| GET | /api/students | Get all students | 200, 500 |
| GET | /api/students/:id | Get single student | 200, 400, 404, 500 |
| GET | /api/students/search/:key word | Search by name/ID | 200, 500 |
| PUT | /api/students/:id | Update student | 200, 400, 404, 500 |
| DELETE | /api/students/:id | Delete student | 200, 400, 404, 500 |

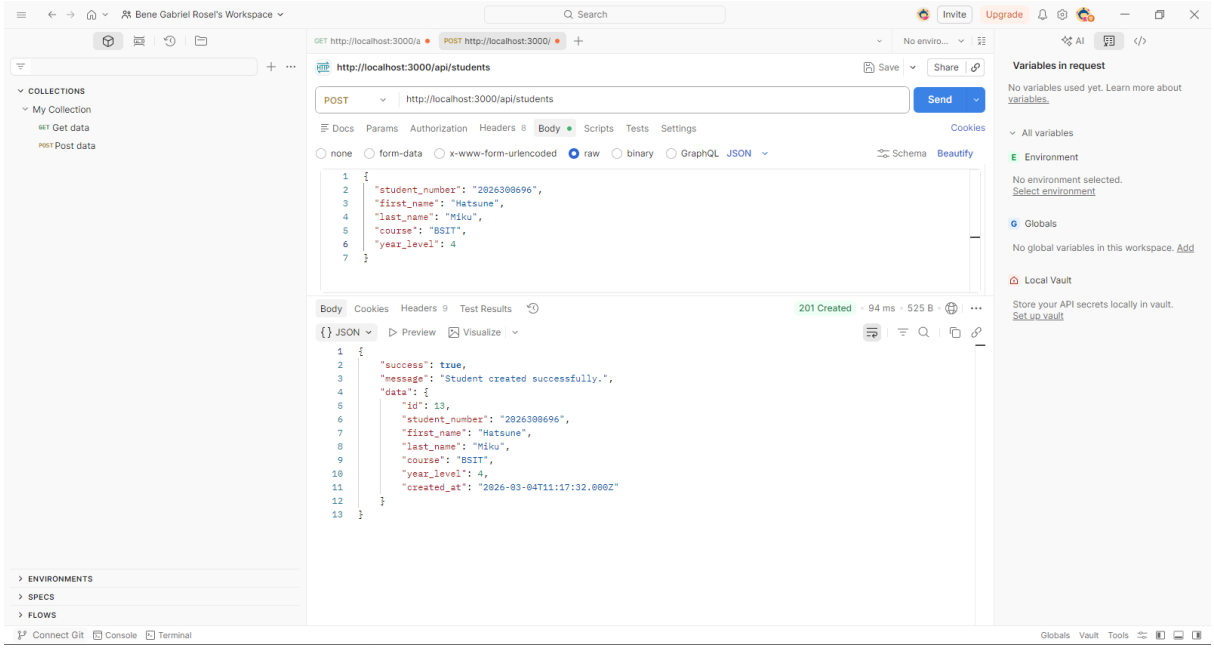
# Screenshots

## • Database Table

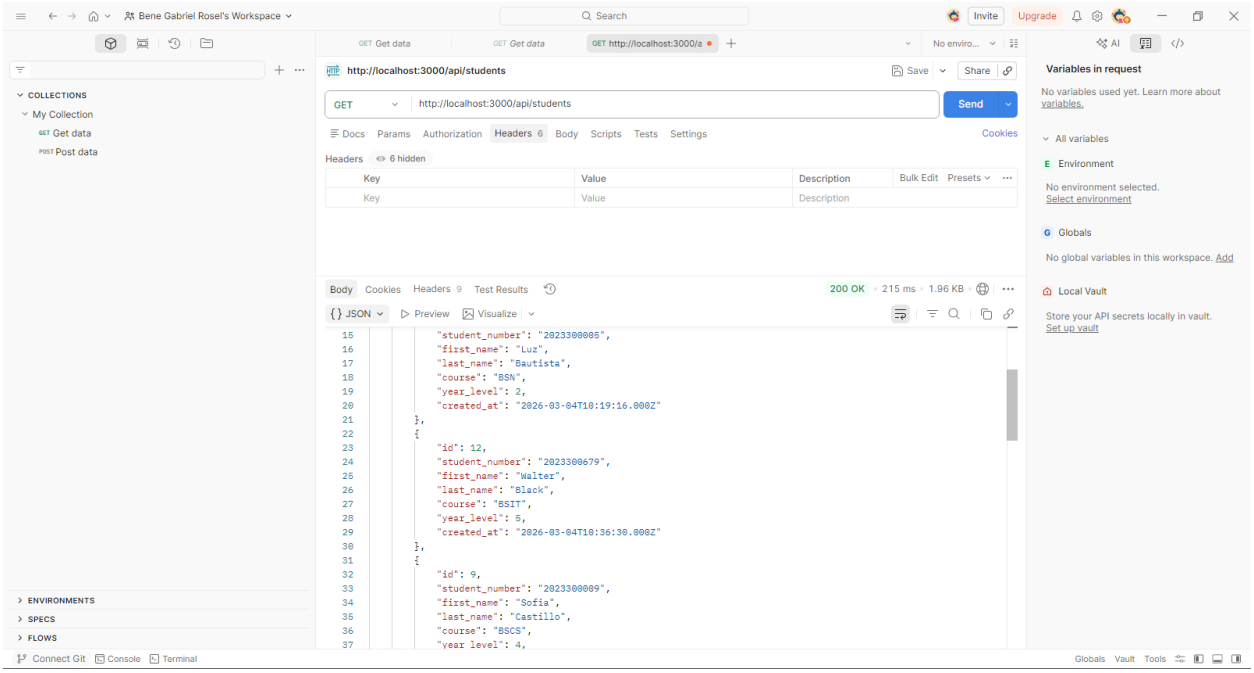



## • Postman Testing

### I. Insert Student (POST)

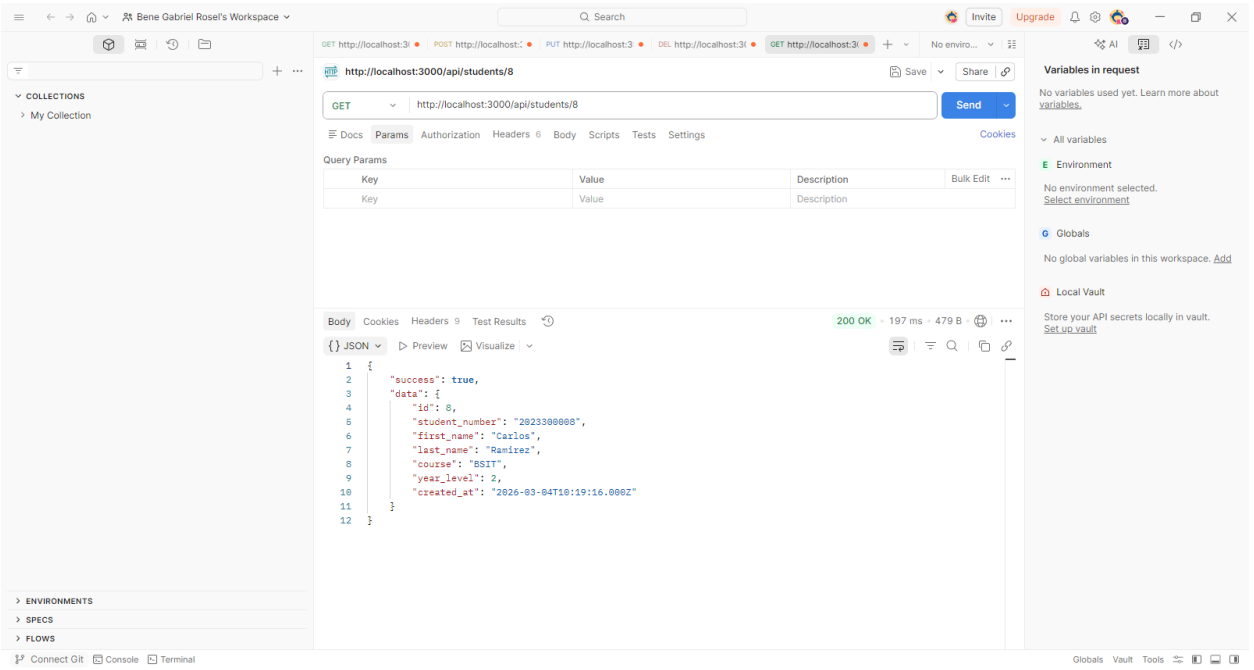


## II. Select All Students (GET)

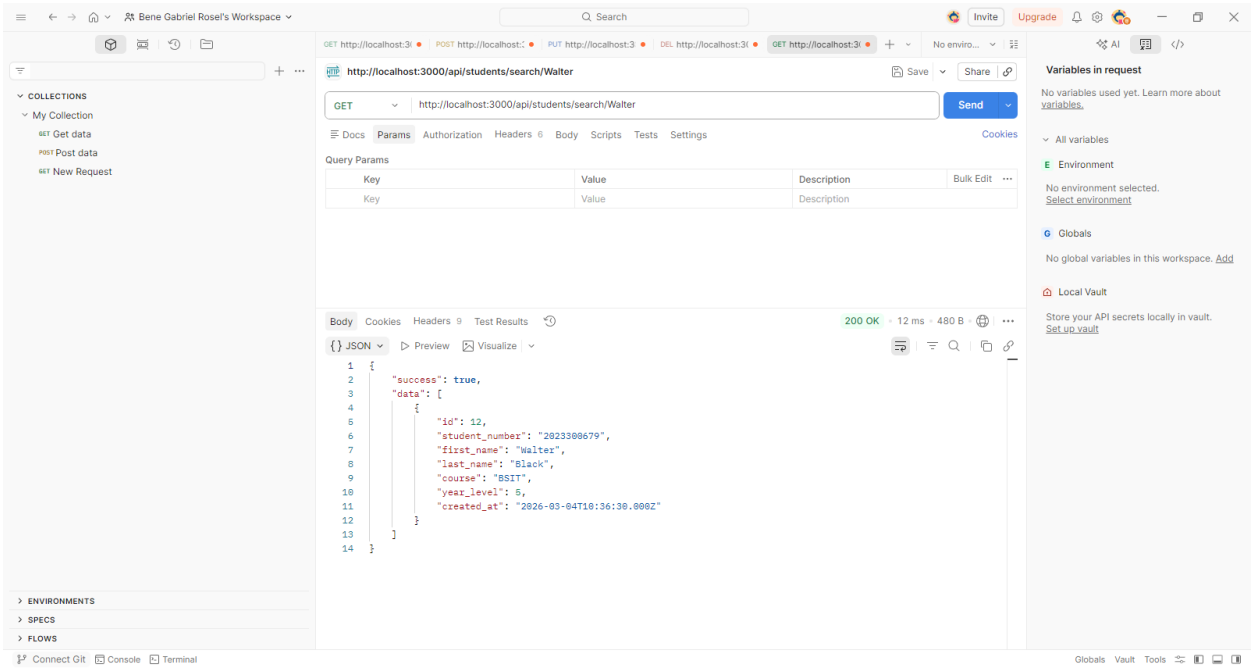



## III. Select Student (GET)

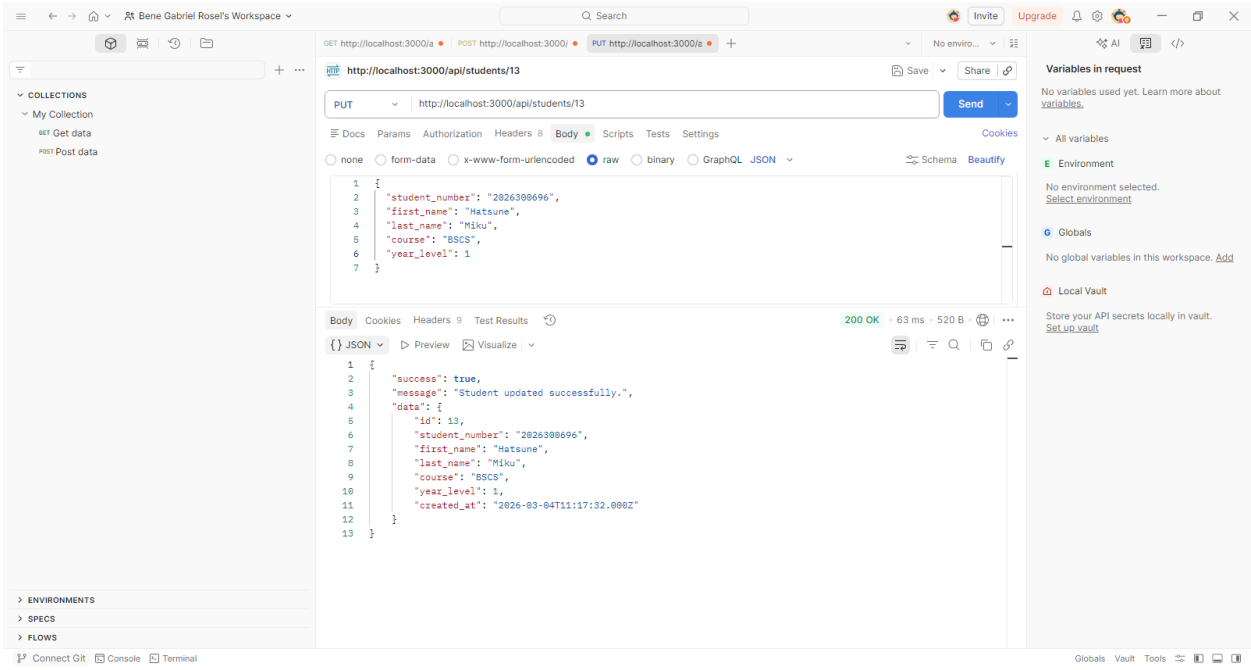
- By ID



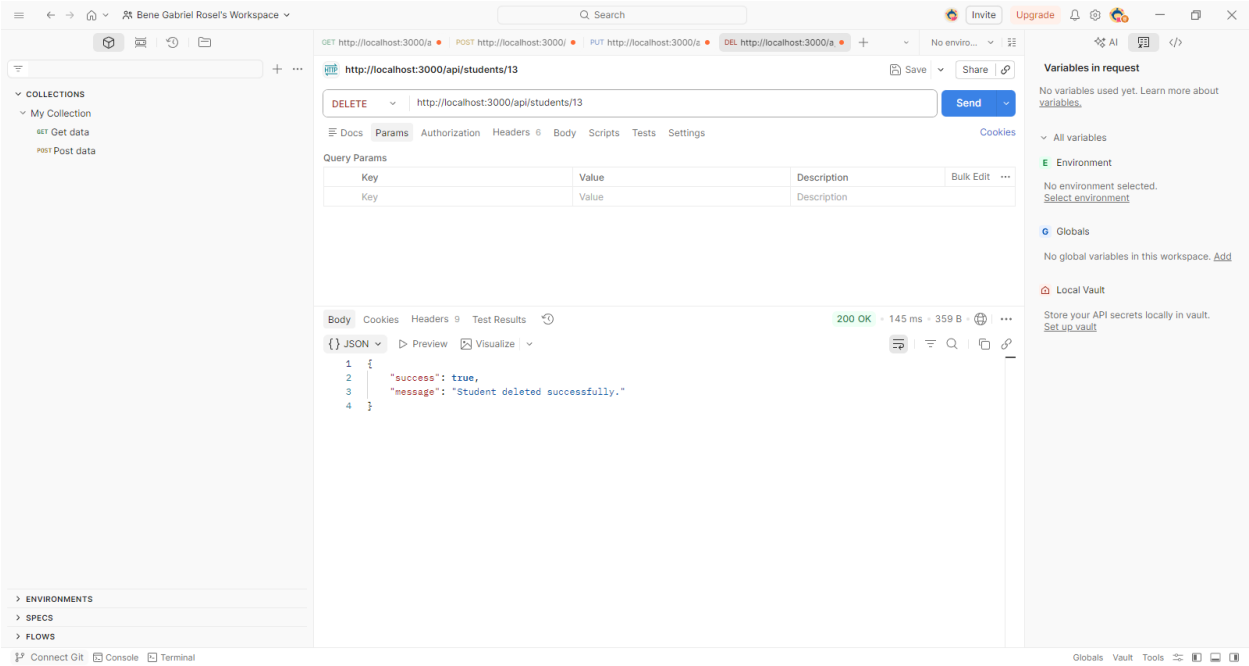
- By name/ID




## IV. Update Student (PUT)



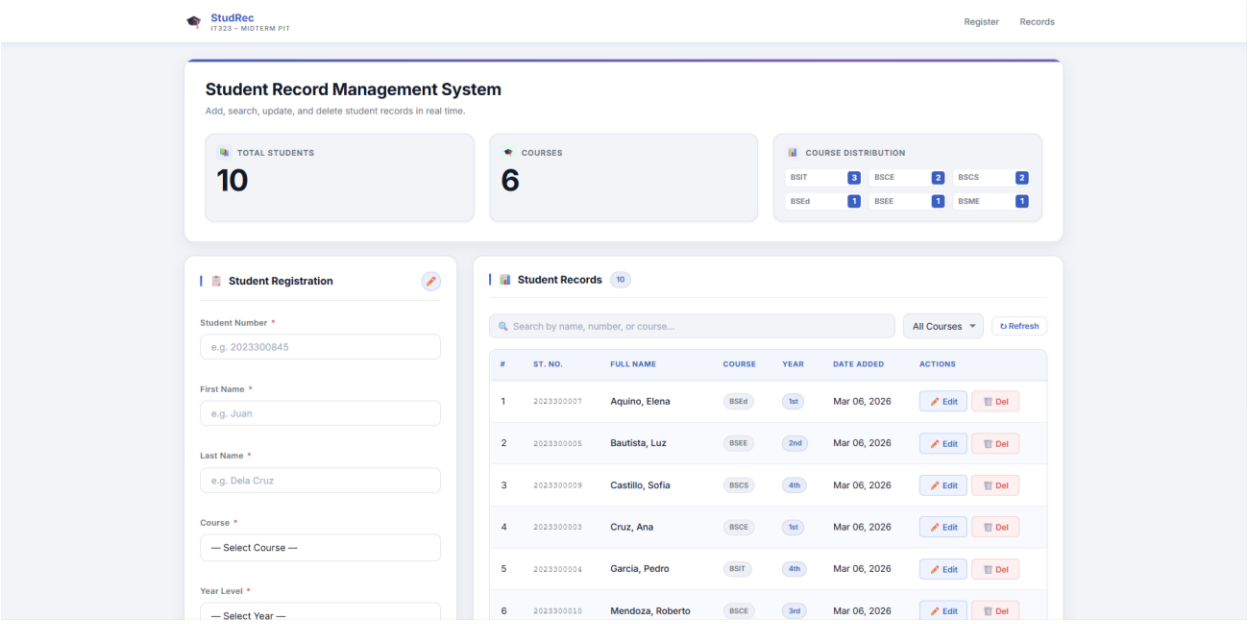
## V. Delete Student (DELETE)



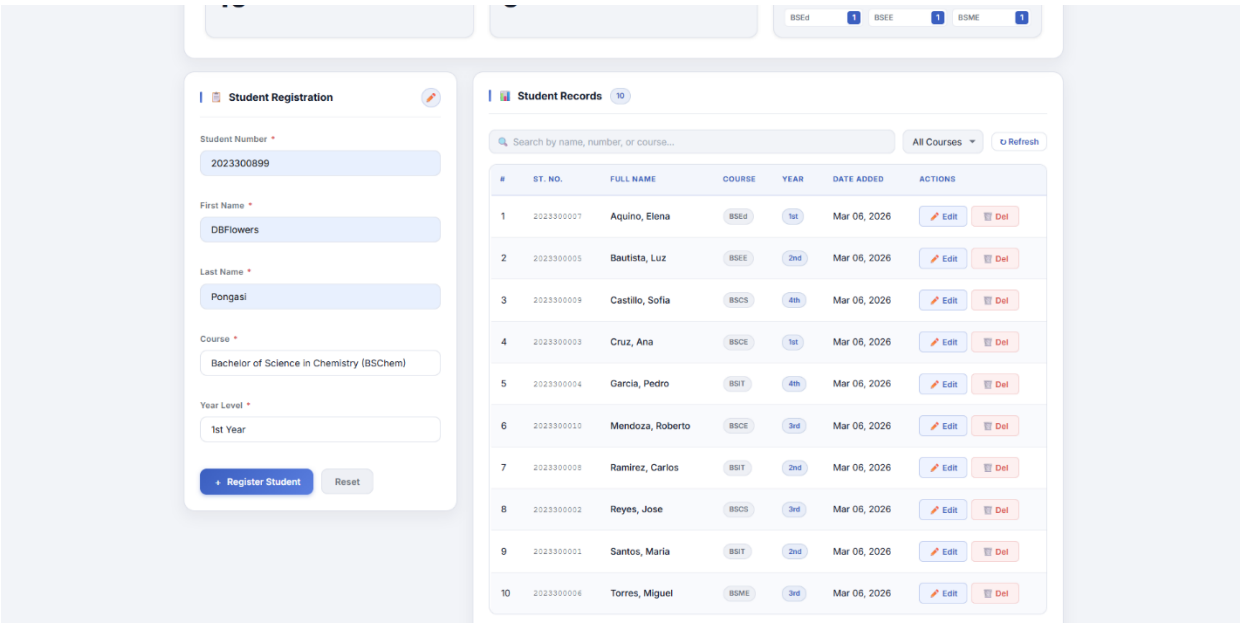

- **Front-End Design and Responsiveness**

## I. Desktop

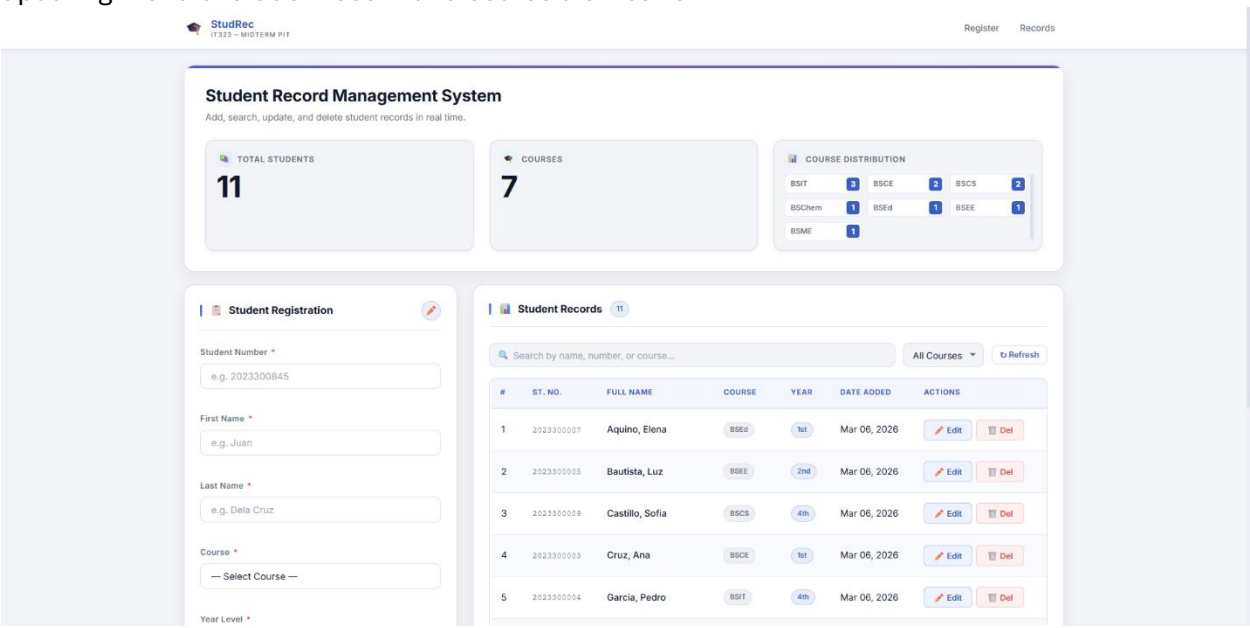
1. Main Dashboard Interface: The main dashboard showing system statistics, the student registration form, and the student records table.



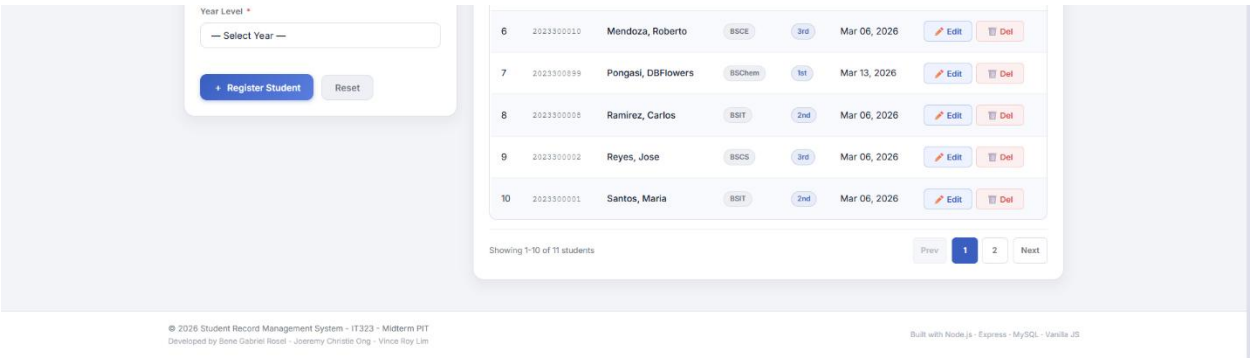
2. Student Registration: The administrator enters the student number, name, course, and year level to register a new student record.



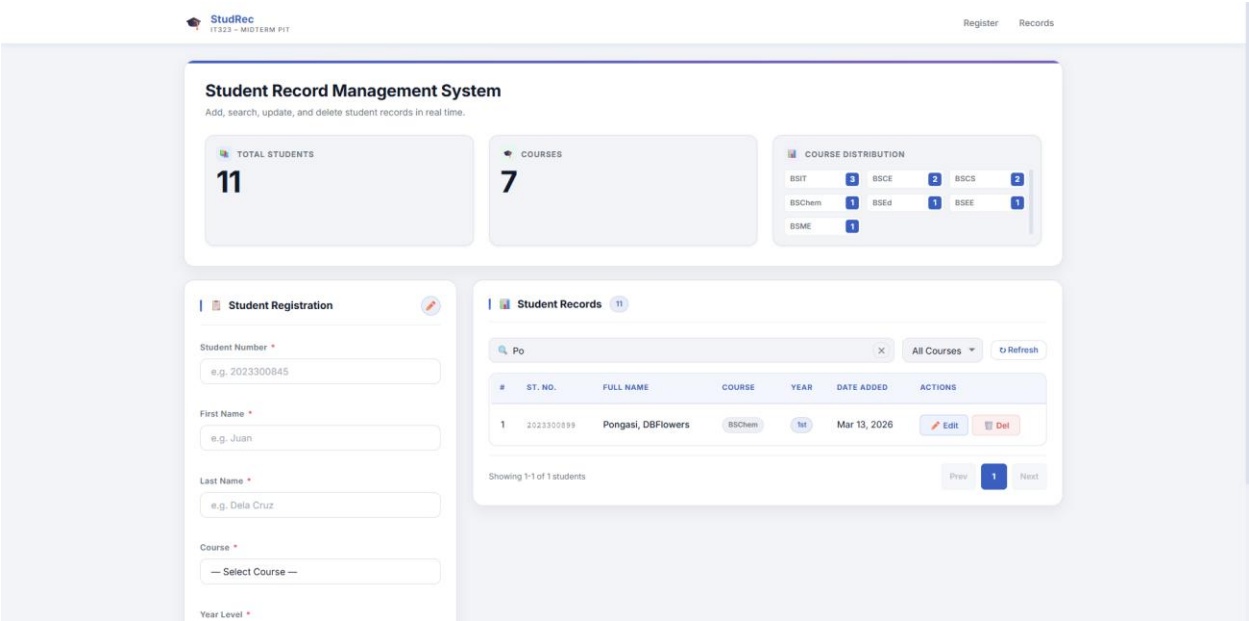

3. After Registering Student: The table automatically refreshes after a new student is added, updating the total student count and course distribution.



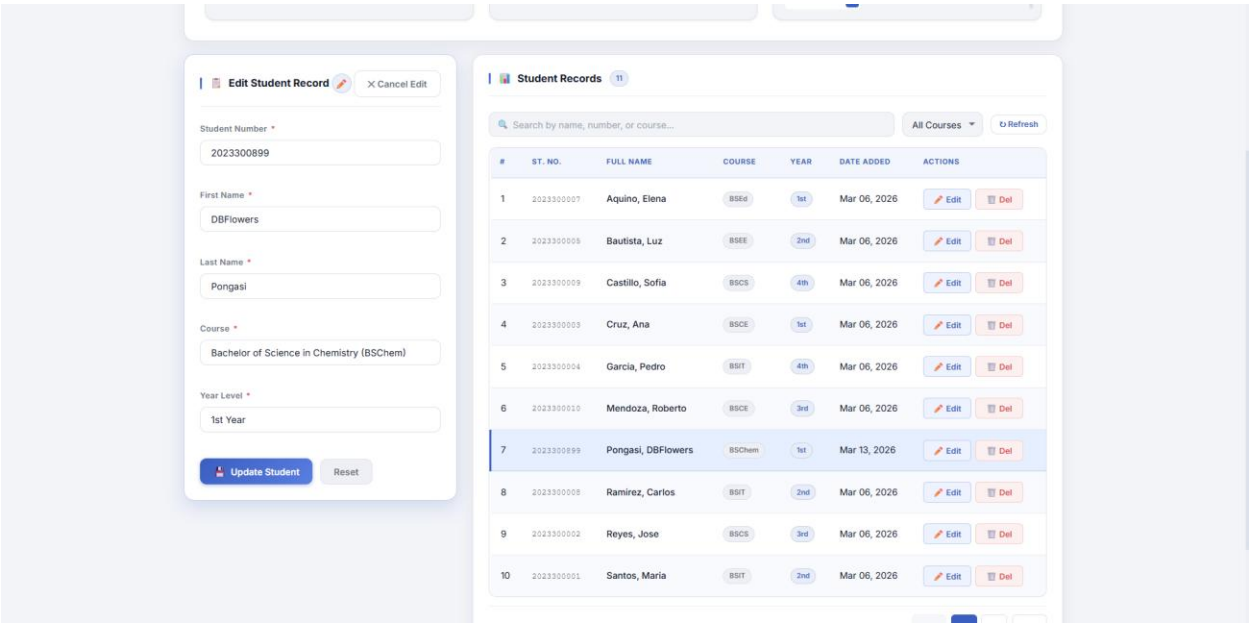

4. Student Records Table with Pagination and Footer: The table supports pagination for easier navigation when multiple student records exist.



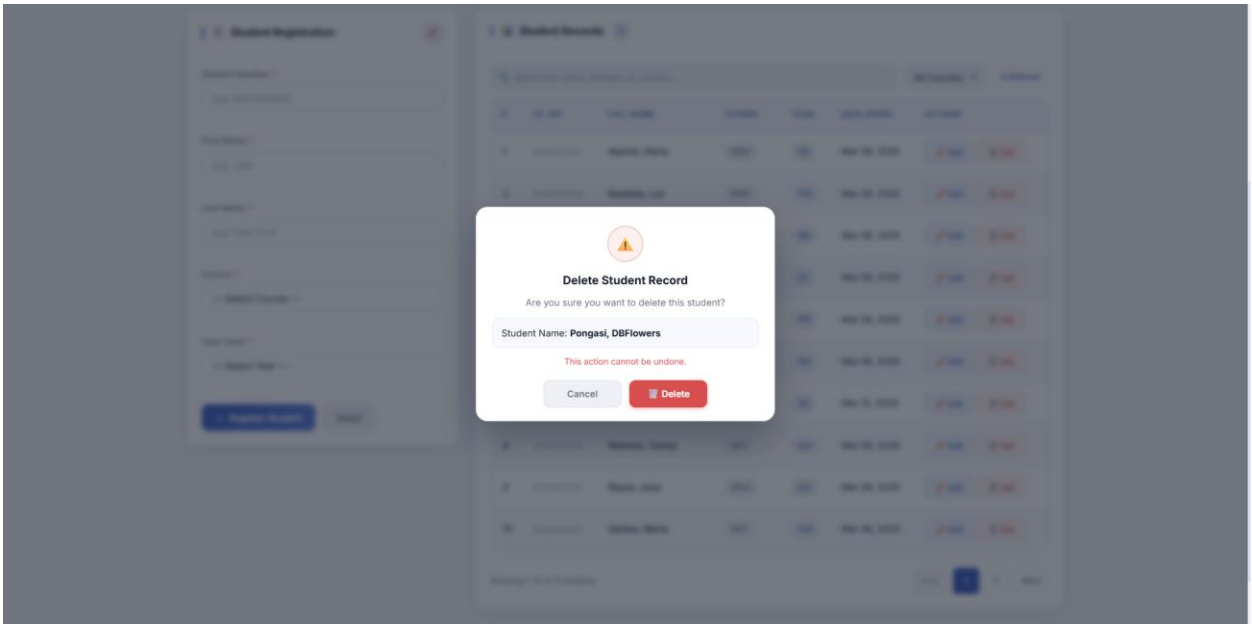
5. Student Search Functionality: The system dynamically filters student records based on name, student number, or course.



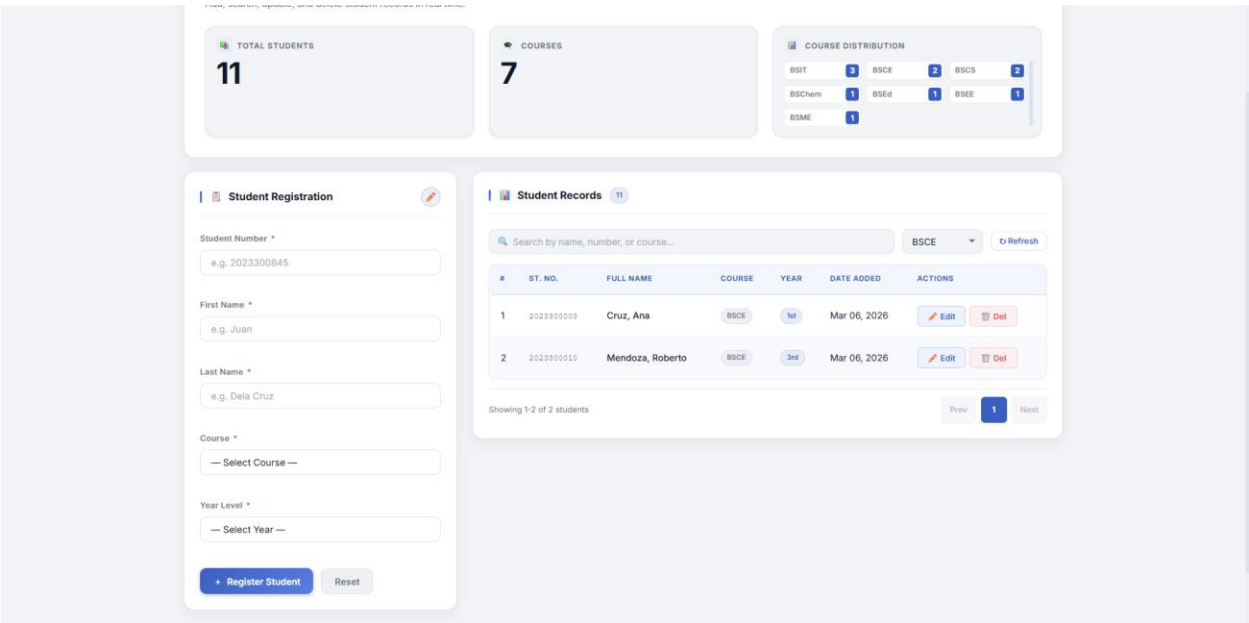

6. Edit Student Record: Clicking the Edit button populates the registration form with the selected student's data, allowing the administrator to update the record.



7. Delete Student Record Confirmation: A confirmation modal appears before deleting a student record to prevent accidental removal.

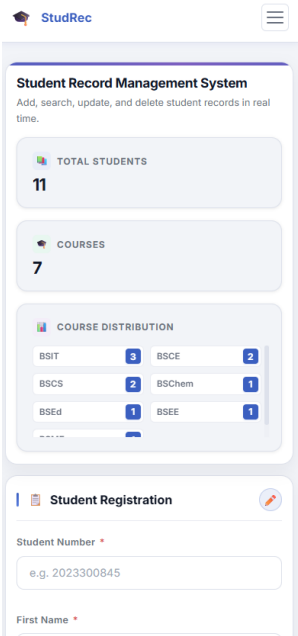



8. Filter Students by Course: Administrators can filter the table by course using the dropdown filter.

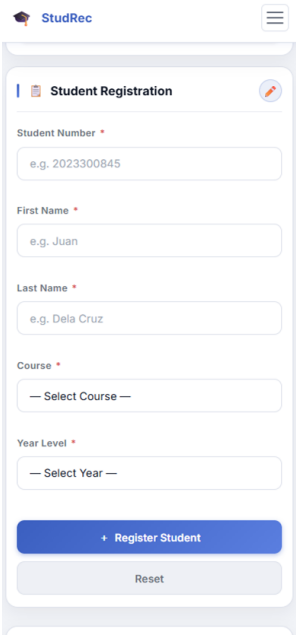


## II. Mobile Responsiveness

1. Dashboard

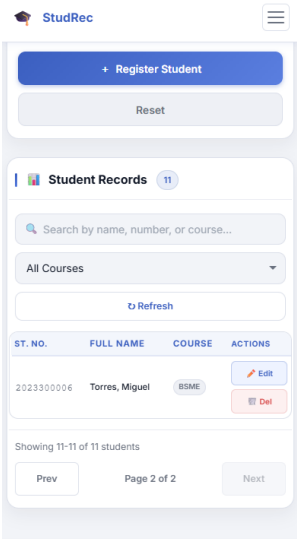



2. Student Registration

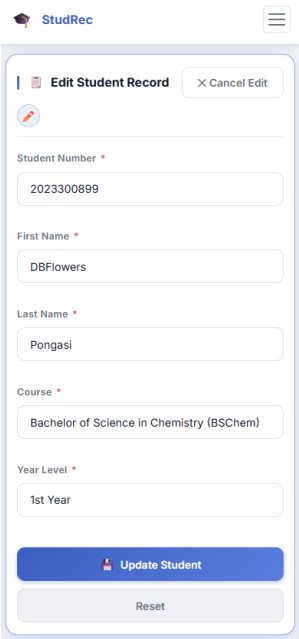



3. Student Records

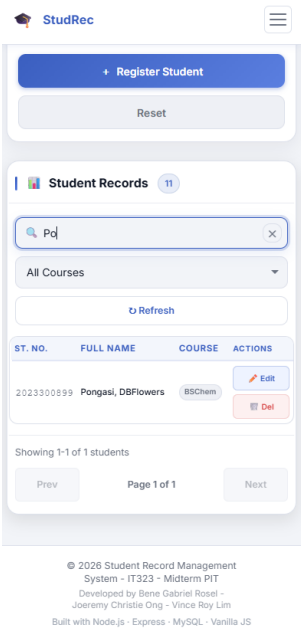
4. Pagination



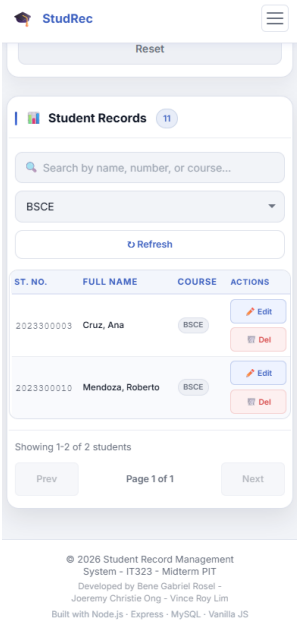

5. Edit Student Record



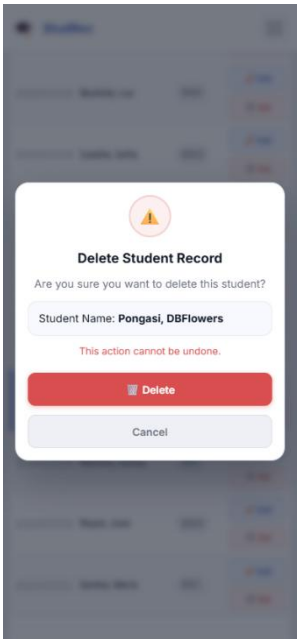

6. Search




7. Filter by Course



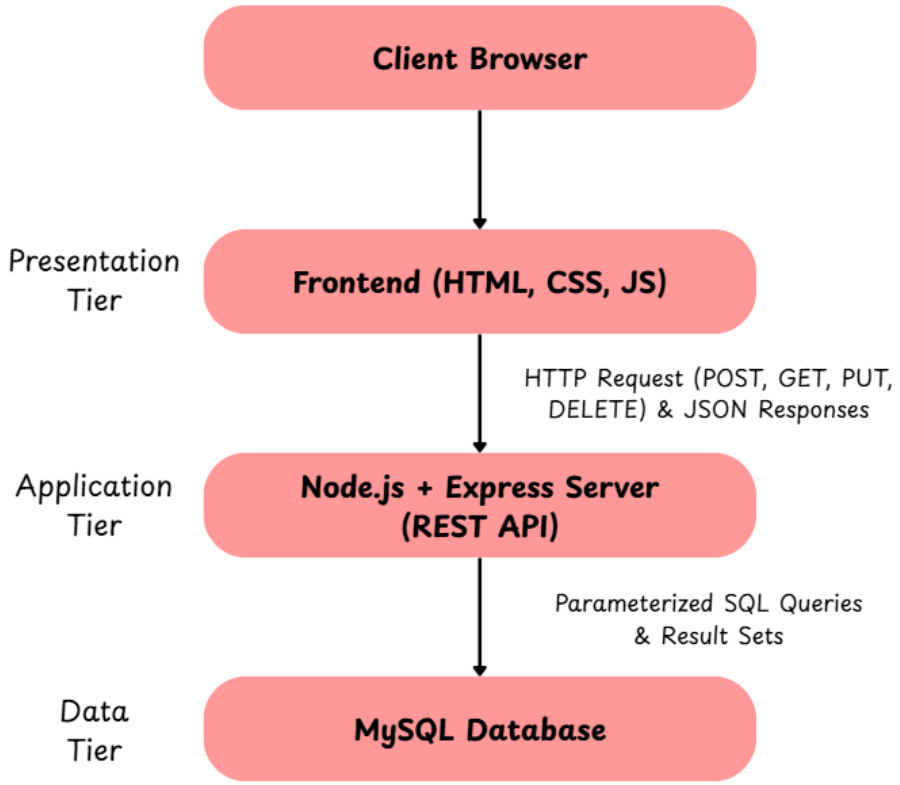

8. Delete Confirmation



## Technical Report

**• System Architecture**

The Student Record Management System (SRMS) is a full-stack web application built following **the three-tier architecture** model:




Project Structure

- server.js - Entry point, mounts middleware + routes
- package.json - Dependencies and npm scripts
- .env.example - Environment variable template (copy to .env)
- config/db.js - MySQL connection pool (mysql2/promise + dotenv)
- routes/students.js - All /api/students route handlers
- public/
  index.html - Single-page frontend
  css/style.css - Full design system (Grid + Flex)
  js/script.js - All Vanilla JS logic
- database/schema.sql - Database schema
- database/seed.sql - Sample data
- docs/IT323-Midterm-PIT-Documentation.pdf - Project documentation

- **Database Design Rationale**

I. Schema Design

| Column | Data Type | Constraints | Description |
|---|---|---|---|
| id | INT | AUTO_INCREMENT PRIMARY KEY | Unique row identifier |
| student_number | VARCHAR(20) | UNIQUE NOT NULL | Official student ID number |
| first_name | VARCHAR(100) | NOT NULL | Student's first name |
| last_name | VARCHAR(100) | NOT NULL | Student's last name |
| course | VARCHAR(100) | NOT NULL | Enrolled Academic Program |
| year_level | INT | NOT NULL CHECK (year_level BETWEEN 1 AND 5), | Current year level (1-4) |
| created_at | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP | Record creation timestamp |

- The AUTO_INCREMENT primary key ensures each record has a unique, system-generated identifier that is safe to use in API routes.
- The UNIQUE constraint on student_number enforces business rules at the database level, preventing duplicate enrollment entries even if application- level validation is bypassed.

II. SQL Operations implemented

| Operation | SQL statement used |
|---|---|
| Insert student | INSERT INTO students (student_number, ...) VALUES (?, ?, ...) |
| Select all student (ordered ASC) | SELECT * FROM students ORDER BY last_name ASC |
| Select student by ID | SELECT * FROM students WHERE id = ? |
| Update student | UPDATE students SET student_number=?, first_name=?, last_name=?, course=?, year_level=? WHERE id=? |
| Delete student | DELETE FROM students WHERE id = ? |
| Search (using LIKE) | SELECT * FROM students WHERE first_name LIKE ? OR last_name LIKE ? OR student_number LIKE ? OR course LIKE ? |

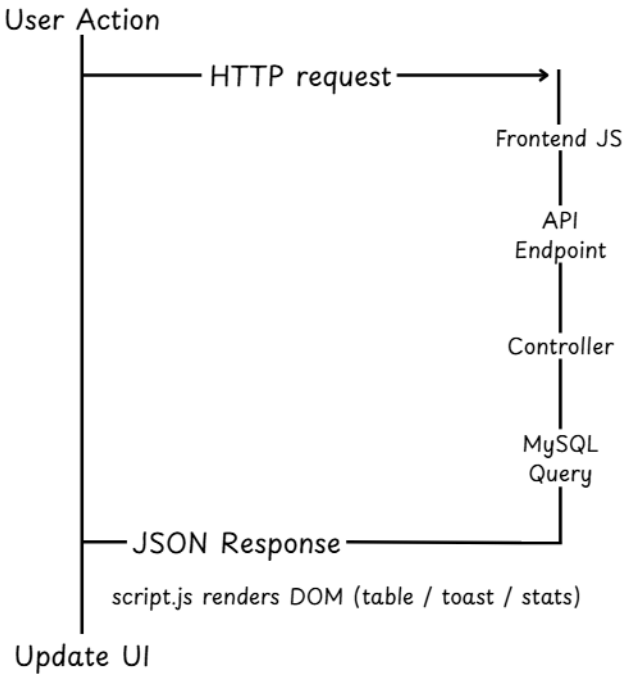
III. Normalization

1NF: All columns are atomic; no repeating groups

2NF: No partial dependencies (single-column PK)

3NF: No transitive dependencies; `course` is a free-text attribute intentionally not normalized into a separate table at this stage

- **API Flow Explanation**




- Error Handling Strategy
  400 - Missing required fields or duplicate `student_number` (caught via `ER_DUP_ENTRY` MySQL error code)
  404 - Row lookup before update/delete; returns early with message
  500 - Unexpected DB errors; caught in `try/catch`, logged server-side, generic message returned to client

- **Challenges Encountered**

- Problem: Express matched `/api/students/search/:keyword` as `/api/students/:id` (interpreting `"search"` as an ID), triggering a MySQL type mismatch error.
  Solution: The `search/:keyword` route was declared **above** the `/:id` route in `routes/students.js`. Express evaluates routes in registration order, so placing the specific literal path first ensures it matches before the generic wildcard.
- Problem: Firing an API call on every keystroke caused excessive server requests and visual flickering.
  Solution: Implemented a 400 ms debounce timer in `script.js` using `clearTimeout` / `setTimeout`. The search API is only called once the user pauses typing, reducing network calls by ~80% during normal use.
- Problem: Rendering student data directly via `innerHTML` with unsanitized data could allow stored XSS if malicious content was inserted into the database.

Solution**:** An `escHtml()` utility function was implemented to escape `&`, `<`, `>`, `"`, and `'` characters before any user-provided data is inserted into the DOM via `innerHTML`.

- Problem: At viewport widths ≤768px, the navigation links overlapped with the brand and badge elements, breaking the layout.
  Solution: A hamburger button (`#nav-toggle`) was added to the navbar. On mobile, the nav links are hidden by default and revealed as a full-width vertical dropdown when the hamburger is clicked. The button animates into a close (✕) icon when the menu is open using CSS-only bar transforms.

- **Future Improvements**

  - User Authentication
    Implement JWT-based authentication with role-based access control (admin vs. registrar) using `jsonwebtoken` and `bcrypt` for password hashing.
  - Normalized Courses Table
    Create a separate `courses` master table with a foreign key from `students.course_id`, enabling course management (add/archive courses) without modifying student records directly.
  - Data Export
    Allow administrators to export the student list as a CSV or PDF report via a `GET /api/students/export` endpoint using `json2csv` or `pdfmake`.
  - Automated Testing
    Add unit tests for route handlers using `Jest` + `supertest`, and integration tests against a dedicated test database to ensure regression safety during development.